

2024 Ph H2 Q12

Section: Particles and Waves

Topic: Refraction & Total Internal Reflection

Summary of Question

A student investigates refraction using a laser, a Perspex block and a protractor.

- (a) How to process measurements of i and r to obtain a reliable refractive index.
- (b) Given $n = 1.50$, calculate the critical angle for Perspex-air.
- (c) A ray inside Perspex meets a Perspex-air boundary at 50° . Complete the path and mark relevant angles.

(a) Processing measurements to find refractive index

Use Snell's law: $n = \sin i / \sin r$ (for air $\rightarrow$ Perspex).

Plot a straight-line graph to reduce random error:

- Option 1: plot $\sin i$ (y-axis) against $\sin r$ (x-axis). The gradient of the best-fit line is n .
- Option 2: plot $\sin r$ against $\sin i$; then $n = 1 / (\text{gradient})$.

Final answer: Plot $\sin i$ vs $\sin r$ and take the gradient as n (use multiple pairs with a best-fit line).

(b) Critical angle for Perspex-air

At the critical angle θ_c , the refracted ray emerges along the boundary: $\sin \theta_c = 1 / n$.

$$\theta_c = \sin^{-1}(1/1.50) = 41.8^\circ.$$

Final answer: 41.8° (to 3 s.f.)

(c) Ray at 50° to the normal inside Perspex (towards air)

Since $50^\circ > \theta_c$ (41.8°), total internal reflection occurs.

Complete the diagram with a reflected ray at 50° to the normal inside the Perspex; no refracted ray in the air.

Final answer: TIR: reflect at 50° (angle of reflection = angle of incidence), no transmitted ray.

Revision Tips

Always process multiple (i , r) pairs via a $\sin i$ vs $\sin r$ plot and take the gradient.

Critical angle only applies when going from higher n to lower n ; $\sin \theta_c = n_2/n_1$ ($\approx 1/n$ for Perspex $\rightarrow$ air).

If $i > \theta_c$, draw only a reflected ray; if $i < \theta_c$, draw both reflected and refracted rays (with refraction away from the normal).